

Program : Diploma in Instrumentation Engineering	
Course Code : 3088	Course Title: Maintenance & Calibration Workshop I
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide hands-on experience with the Safety and 5S to be followed in instrument workshop
- To familiarize basic measurement techniques of electrical parameters
- To assemble and test electrical and electronic circuits as per given circuit diagram

Course Prerequisites:

Topic	Course code	Course name	Semester
DC & AC electric circuits, series and parallel circuits, voltage and current measurement, active and passive electronic components		Fundamentals of electrical and electronics	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Implement 5S and follow safety measures in lab	6	Applying
CO2	Practice electrical parameters measurement techniques	15	Applying

CO3	Demonstrate the wiring of electrical circuits and take reading of electrical parameters	12	Applying
CO4	Implement electronic circuits on PCB	9	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				2	2	3
CO2	3			2	2	2	
CO3	3			2			
CO4	3	2	2	2			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Implement 5S and follow safety measures in lab		
M1.1	5S Technique Implement 5S to be followed in Instrument workshop.	3	Applying
M1.2	Industrial Safety i) Understand safety measures to be followed in industrial environment. ii) Study and Identify the equipments used for fire & safety Risk Management.	3	Applying

CO2	Practice electrical parameters measurement techniques.		
M2.1	Voltage and current measurement of DC circuit Measure voltage and current of a DC circuit by selecting proper instrument and range, reading of meter in correct position and correct way by using multi range meter.	3	Applying
M2.2	Voltage, current and resistance measurement using Multimeter i) Measure the voltage and current of AC & DC circuits using a multimeter. ii) Measure the values of carbon resistors using multimeter and compare these values with the colour code	1.5	Applying
M2.3	Measuring size of cables: Measure the size of various cables using standard wire gauge and micrometer.	3	Applying
M2.4	Status of switches using multimeter or test lamp Verify the status of switches (SPST, SPDT, DPST, DPDT, TPST, TPDT, Push button switches, Rotary switches, Pull switch) using multimeter or test lamp.	1.5	Applying
M2.5	Maintenance of lead acid battery Practice maintenance of lead acid battery.	3	Applying
M2.6	UPS Circuits: Familiarize UPS circuits and practice trouble shooting works	3	Applying
	Lab Test – I	1	
CO3	Demonstrate the wiring of electrical circuits and take reading of electrical parameters		
M3.1	Series and Parallel Electric Circuits Assemble the following circuits and verify voltage and current divisions: i) Series circuit contains three lamps of different power ratings. ii) Parallel circuit contains three lamps of different power ratings.	3	Applying

M3.2	Wattmeter and Energy meter i) Read a wattmeter connected in an existing load. ii) Read an energy meter connected in an existing installation	3	Applying
M3.3	Electric Wiring Carry out rigid PVC surface conduit wiring to control: a) One lamp by one switch. b) Two lamps and a socket outlet by independent switches. c) One lamp from two different places (stair case) by two switches. d) One lamp from multiple places using intermediate switch (more than two places). e) Lamps from different places (Godown wiring). f) Lamps from different places independently and from one specific place by a Master switch. g) Lamps from different places in series-parallel (Hospital wiring).	6	Applying
CO4	Implement electronic circuits on PCB		
M4.1	Soldering Practice i) Solder and desolder electronic components on PCB. ii) Implement electronic circuits on PCB and test the working (Half wave rectifier, Full wave rectifier, Astable Multivibrator, Power supply unit regulated at 12V etc)	9	Applying
	Lab Test – II	2	

Text / Reference:

T/R	Book Title/Author
T1	M L Gupta, Workshop Practice in Electrical Engineering for Industrial Training Institutes, Technical Schools, Polytechnics & Electrical Supervisory Examinations
R1	Bhawani Shankar Chowdhry, The First Book of Electronics Workshop, River Publishers
R2	S K Bhattacharya, Basic Electrical and Electronics Engineering, Pearson Publications

Online Resources:

Sl. No	Website Link
1	https://www.youtube.com/watch?v=rzCwptOFSC8I
2	https://www.youtube.com/watch?v=QGkyYkx2NFY
3	https://www.youtube.com/watch?v=4gR_osFn9FE
4	https://www.makerspaces.com/how-to-solder/
5	https://www.youtube.com/watch?v=Qps9woUGkVI